

ISTQB QA/QC Role Mind Map

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mindmap
  root((ISTQB QA/QC Roles))
    QA (Quality Assurance)
      Process Improvement
        Define Standards
        Establish Procedures
        Audit Processes
        Continuous Improvement
      Quality Planning
        Test Strategy
        Quality Objectives
        Risk Management
      Prevention Focus
        Defect Prevention
        Process Optimization
      Compliance
        ISO Standards
        Regulatory Requirements
      Test Automation
        Automation Strategy
        Framework Selection
        CI/CD Integration
        Tool Maintenance
    QC (Quality Control)
      Product Verification
        Test Execution
        Defect Detection
        Validation Activities
      Inspection
        Reviews
        Walkthroughs
        Inspections
      Testing Activities
        Functional Testing
        Non-Functional Testing
          Performance Testing
          Security Testing
          Usability Testing
        Regression Testing
        Retesting
```

Test Analyst

Test Design

- Test Cases

- Test Data

- Test Conditions

Test Execution

- Manual Testing

- Automated Testing Support

- Defect Detection

- Defect Reporting

- Traceability

Test Automation Engineer

Automation Framework

- Test Script Development

- Keyword-Driven Framework

- Data-Driven Framework

- CI/CD Integration

- Tool Maintenance

- Reporting & Metrics

Test Manager

- Test Planning

- Resource Management

- Estimation

- Risk Management

- Stakeholder Communication

- Test Monitoring

- Test Closure

QA/QC Skills

- Analytical Thinking

- Communication

- Attention to Detail

- Domain Knowledge

- Tool Knowledge

ISTQB Core Concepts

- Testing Principles

- Test Levels

- Unit Testing

- Integration Testing

- System Testing

- Acceptance Testing

- Test Types

- Functional Testing

- Non-Functional Testing

- Performance Testing

- Security Testing

- Usability Testing

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graph TD; QA_QC_Mindmap[QA_QC_Mindmap] --> Compatibility_Testing[Compatibility Testing]; QA_QC_Mindmap --> Testing_Techniques[Testing Techniques]; QA_QC_Mindmap --> Tools[Tools]; QA_QC_Mindmap --> Performance_Testing[Performance Testing]; QA_QC_Mindmap --> Security_Testing[Security Testing]; Compatibility_Testing --> Static_Testing[Static Testing]; Compatibility_Testing --> Dynamic_Testing[Dynamic Testing]; Compatibility_Testing --> Defect_Lifecycle[Defect Lifecycle]; Compatibility_Testing --> Risk_Based_Testing[Risk-Based Testing]; Static_Testing --> Reviews[Reviews]; Static_Testing --> Static_Analysis[Static Analysis]; Dynamic_Testing --> Black_Box_Testing[Black-Box Testing]; Dynamic_Testing --> White_Box_Testing[White-Box Testing]; Tools --> Test_Management[Test Management]; Tools --> Automation[Automation]; Test_Management --> JIRA[JIRA]; Test_Management --> TestRail[TestRail]; Test_Management --> Zephyr[Zephyr]; Automation --> Selenium[Selenium]; Automation --> Cypress[Cypress]; Automation --> Playwright[Playwright]; Automation --> Appium[Appium]; Performance_Testing --> JMeter[JMeter]; Performance_Testing --> LoadRunner[LoadRunner]; Performance_Testing --> Gatling[Gatling]; Security_Testing --> Burp_Suite[Burp Suite]; Security_Testing --> OWASP_ZAP[OWASP ZAP];
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- Compatibility Testing
 - Static Testing
 - Reviews
 - Static Analysis
 - Dynamic Testing
 - Black-Box Testing
 - White-Box Testing
 - Defect Lifecycle
 - Risk-Based Testing
- Tools
 - Test Management
 - JIRA
 - TestRail
 - Zephyr
 - Automation
 - Selenium
 - Cypress
 - Playwright
 - Appium
- API Testing
 - Postman
 - REST Assured
 - SoapUI
- Performance Testing
 - JMeter
 - LoadRunner
 - Gatling
- Security Testing
 - Burp Suite
 - OWASP ZAP